Lecture Notes on RECURSIVE ARITHMETIC
by
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Preface. The purpose of these notes is to give a rough outline of the material presented in the Recursive Arithmetic Seminar by J. R. Guard, during 1962-63 at the Mathematics Department, Princeton University.

Chapter 1. Primitive Recursive Arithmetic

In these notes we shall refer to the non-negative integers as *natural numbers* or *integers*. By *function* - unless otherwise noted - we shall mean a function from natural numbers to natural numbers.

We shall define the notion of a primitive recrusive function informally. Later we shall explicate this notion within the framework of a formal system.

DEFINITION 1. The notion of a *primitive recursive* (p.r.) *function* is defined inductively by

a) a function f satisfying one of the equations

$$f(x) = 0$$

$$f(x) = x + 1$$

$$f(x_1,\ldots,x_n) = x_i \quad (1 \le i \le n;\ n = 1,2,\ldots)$$

is a primitive recursive function,

b) if $g, h_1, \ldots, h_m$ are primitive recursive functions then f is a primitive recursive function where

$$f(x_1,\ldots,x_n) = g(h_1(x_1,\ldots,x_n),\ldots,h_m(x_1,\ldots,x_n))$$
$$(m = 1,2,\ldots;\quad n = 1,2,\ldots)$$

c) if g and h are primitive recursive functions then f is a primitive recursive function where

$$f(x_1,\ldots,x_n,0) = g(x_1,\ldots,x_n)$$

$$f(x_1,\ldots,x_n,\ x_{n+1} + 1) = h(x_1,\ldots,x_{n+1},f(x_1,\ldots,x_{n+1}))$$
$$(n = 1,2,\ldots)\ .$$

EXERCISE 1. Show that the following functions are p.r.:

[1] $x + y$ [2] $x \cdot y$ [3] x^y

[4] $x - y = \begin{cases} x - y, & \text{if } x \geq y \\ 0, & \text{otherwise} \end{cases}$

[5] $\max(x_1,\ldots,x_n)$ [6] p_x (the $x+1$ -th prime)

[7] $[x/y]$ (the greatest integer $\leq x/y$)

[8] $(x)_y$ (the number of times p_y is a factor of x)

EXERCISE 2. Show that for each p.r. function g, there are p.r. functions Σ_g and Π_g satisfying

$$\Sigma_g(x_1,\ldots,x_n,y) = \sum_{i=0}^{y} g(x_1,\ldots,x_n,i)\,, \quad \text{and}$$

$$\Pi_g(x_1,\ldots,x_n,y) = \prod_{i=0}^{y} g(x_1,\ldots,x_n,i)\,.$$

DEFINITION 2. Notations - called <u>primitive recursive functors</u> - for the p.r. functions are defined inductively by

a) $s, \theta, u_1^{(1)}, u_1^{(2)}, u_2^{(2)}, u_1^{(3)},\ldots$ are respectively functors naming the p.r. functions as defined in Definition 1 a).

b) if $G, H_1,\ldots,H_m$ are p.r. functors naming the p.r. functions $g, h_1,\ldots,h_m$ respectively then $CGH_1H_2\ldots H_m$ is the functor naming f as defined in Definition 1 b), and

c) if G and H are p.r. functors naming g and h then RGH is the functor naming f as defined in Definition 1 c).

Notice that a given p.r. function will have many (in fact denumerably - many) functors which name it. To each functor we can uniquely assign a positive integer n which corresponds to the number of arguments the function named by the functor can take. We call this functor an <u>n-ary</u> functor.

EXERCISE 3. Find a 2-ary functor naming addition.

DEFINITION 3. The notions of <u>primitive recursive term</u> and <u>primitive recursive formula</u> are defined inductively by

a) a natural number is a p.r. term,

b) a variable which ranges over the natural numbers is a p.r. term,

c) if $A_1,\dots,A_n$ are p.r. terms and F is an n-ary p.r. functor then $F(A_1,\dots,A_n)$ is a p.r. term,

d) if A and B are p.r. terms then $A = B$ is a p.r. formula, and

e) if P and Q are p.r. formulas then $\sim P$, $(P \supset Q)$, $(P \vee Q)$, $(P \& Q)$, and $(P \equiv Q)$ are p.r. formulas.

P.r. formulas and terms which do not contain variables are called <u>variable-free</u>.

THEOREM 1. For every variable-free p.r. term (vft) A, there is an integer n, called the <u>value</u> of A, such that $A = n$ is true. There is an algorithm for computing the value of any vft.

EXERCISE 4. Find a reasonable algorithm for computing vfts.

EXERCISE 5. If A is a p.r. term, all of whose variables appear in the list $x_1,x_2,\dots,x_n$ then show that there is an n-ary p.r. functor F_A such that

$$F_A(x_1,\dots,x_n) = A \ .$$

EXERCISE 6. If P is a p.r. formula, all of whose variables appear in the list $x_1,x_2,\dots,x_n$ then show that there is an n-ary p.r. functor F_P such that

$$F_P(x_1,\dots,x_n) \leq 1 \quad \&(F_P(x_1,\dots,x_n) = 0 \equiv P) \ .$$

Use Exercise 2 and this exercise to define the notions of bounded existential and existential quantification, as well as a bounded least-number operator.

EXERCISE 6. (Continued) Show by using the result of this exercise that definition by cases is a p.r. process.

DEFINITION 4. Let f be any 1-ary p.r. function. We write $f^y(x)$ for the p.r. function $g(x,y)$ defined by

$$g(x,o) = x$$

$$g(x,s(y)) = f(g(x,y)).$$

If A and B are p.r. terms, then

$$F^A(B) \rightarrow Ru_1^{(1)}CFu_3^{(3)}(B,A)$$

for a 1-ary p.r. functor F . (" $\rightarrow$ " is read "is an abbreviation for."). Also

$$F\ A \rightarrow F(A)$$

THEOREM 2. The schema

$$f(x,o) = g(x)$$

$$f(x,sy) = h(x,y,f(k(x),\ y))$$

is p.r. I.e. if g, h, and k are p.r., then f is p.r.

Proof: Define τ p.r. ly by

$$\tau(x,\ y,\ o) = g(k^y(x))$$

$$\tau(x,\ y,\ s\ s) = h(k^{y \dot{-} s\ s}(x),\ s,\ \tau(x,y,s)).$$

By induction on s we can show

$$y \geq s \supset \tau(x,\ sy,\ s) = \tau(k(x),\ y,\ s)$$

and

$$f(x,\ s) = \tau(x,\ s,\ s).$$

EXERCISE 7. Show how Theorem 2 can be extended to prove that

$$f(x,\ o) = g(x,o)$$

$$f(x,\ sy) = h(x,\ y,\ f(k(x,y),\ y))$$

and more generally

$$f(x,\ o) = g(x,\ o)$$

$$f(x,\ sy) = h(x,y,f(k_1(x,y),y),\ldots,f(k_m(x,y),y))$$

are p.r. schema. These results can be extended to analogous schema defining n-ary functions .

We wish to show that "course-of-value recursion" is a p.r. process.

Consider the schema

$$f(x,\ o) = g(x)$$

$$f(x,\ sy) = h(x,y,f(k(x,y),\ e(y)))$$

where $e(y) < sy$.

This schema is of a radically different nature, for we have no effective method of determining whether a given schema is of this type. (determining the truth of $e(y) < sy$ could be highly difficult).

We can circumvent this obstacle by the following device.

DEFINITION 5. For a 1-ary p.r. function e , let $[e,y]$ be a p.r. term satisfying

$$[e,y] = \begin{cases} e(y), & \text{if } e(y) \leq y \\ o \ , & \text{otherwise.} \end{cases}$$

THEOREM 3. The schema (called a <u>course-of-values recursion</u>)

$$f(x,o) = g(x)$$

$$f(x,\ sy) = h(x,y,f(k(x,y)\ ,\ [e,y])) \quad \text{is p.r.}$$

Proof: It suffices to define a p.r. function Φ satisfying

$$\Phi(x,y) = 2^{f(x,o)} 3^{f(x,1)} \cdots\cdots p_y^{f(x,y)}$$

THEOREM 4. The schema (called a <u>simultaneous recursion</u>)

$$f_o(x,o) = g_o(x)$$

$$f_1(x,o) = g_1(x)$$

$$\text{—— —— —}$$

$$f_m(x,o) = g_m(x)$$

$$f_o(x,sy) = h_o(x,y,f_o(x,y),\ldots,f_m(x,y))$$

$$\text{—— —— —— ——}$$

$$f_m(x,sy) = h_m(x,y,f_o(x,y),\ldots,f_m(x,y))$$

is p.r.

Proof: It suffices to define a p.r. function Φ satisfying

$$\Phi(x,y) = 2^{f_0(x,y)}\, 3^{f_1(x,y)} \cdots\cdots p_m^{f_m(x,y)} .$$

THEOREM 5. The schema (called <u>unnested</u> 2-fold <u>recursion</u>)

$$f(x,y) = g(x,y) \quad \text{if} \quad x \cdot y = 0$$

$$f(sx,sy) = h(x,y,f(x,sy),\ f(k(x,y),y))$$

is p.r.

Proof: Let α, ξ, ϵ, ψ be p.r. functions satisfying

$$\alpha(x,y) = \max\{k(i,y) \ : 0 \leq i \leq x\}$$

$$\xi(x,y,s) = g(x,y) \quad , \text{ if } \ x \cdot y = 0$$

$$\xi(sx,sy,s) = h(x,y,\xi(x,sy,s),\ (s)_{k(x,y)})$$

$$\varepsilon(x,y,s) = 2^{\xi(o,y,s)} \cdots\ p_x^{\xi(x,y,s)}$$

$$\psi(x,y) = 0 \quad \text{if} \quad x \cdot y = 0$$

$$\psi(sx,sy) = \varepsilon(\alpha(x,y),\ y\ ,\ \psi(\alpha(x,y),\ y).$$

Then by double induction on x and y we have

$$s \geq x \supset f(x,y) = \xi(x,y,\psi(s,y)).$$

Hence $\qquad f(x,y) = \xi(x,y,\psi(x,y))$

EXERCISE 8. Define for a fixed integer n, a notion of unnested n-fold recursion.

DEFINITION 6. Let J, K, and L be p.r. functions satisfying

$$J(x,y) = (1+2+\cdots+(x+y)) + x \quad y$$

$$K(J(x,y)) = x \quad , \qquad \text{and}$$

$$L(J(x,y)) = y.$$

Note that $\qquad J(Kx,Lx) = x.$

THEOREM 6. From J, K, L, s, o, $u_1^{(1)}$, $u_1^{(2)}$, $u_2^{(2)}$, ... all the p.r. functions can be generated by composition and the recursion schema

$$f(x,o) = g(x)$$
$$f(x,sy) = h(y,f(x,y)) \quad .$$

EXERCISE 9. Prove Theorem 6.

EXERCISE 10. Show that the recursion schema of Theorem 6 could be replaced by

$$f(x,o) = x$$
$$f(x,sy) = h(y,f(x,y)) \quad .$$

EXERCISE 11. Show that from $+$, $\dot{-}$, s , $u_1^{(2)}$, $u_2^{(2)}$, composition, and the recursion schema

$$f(0) = 0$$
$$f(sx) = g(x,f(x))$$

that one can define the functions x^2 , $[x/2]$, $x.y$, and $[\sqrt{x}]$.

EXERCISE 12. Define the "pseudo-pairing" functions J_o , K_o , and L_o by

$$L_o(x) = x \dot{-} [\sqrt{x}]^2$$
$$K_o(x) = L_o([\sqrt{x}])$$
$$J_o(x,y) = ((x+y)^2 + x)^2+y) \quad .$$

Prove that

$$K_oJ_o(x,y) = x \quad , \quad L_oJ_o(x,y) = y \quad , \text{ and}$$
$$L_o(sx) \neq o \supset . \ L_o(sx) \dot{-} 1 = L_o(x) \ \& \ K_o(sx) = K_o(x)$$

EXERCISE 13. (Rosza Péter) show that all the p.r. functions can be generated from $+$ $\dot{-}$, s , $u_1^{(2)}$, $u_2^{(2)}$, $u_1^{(3)}$,...,by applications of composition and the recursion

$$f(0) = 0$$
$$f(sx) = g(x,f(x)) \quad .$$

EXERCISE 14. (Rosza Péter) Show that all the p.r. functions can be generated from $+$, L_o , s , $u_1^{(2)}$, $u_2^{(2)}$, $u_1^{(3)}$,..., by composition and the recursion

$$f(0) = 0$$

$$f(sx) = g(f(x)) \quad .$$

EXERCISE 15. (Rosza Péter) Show that all the 1-ary p.r. functions can be generated from L_o , s by the composition schema

$$f(x) = g(h(x)) \quad ,$$

the special composition schema

$$f(x) = g(x)+h(x)$$

and the recursion schema

$$f(0) = 0$$

$$f(sx) = g(f(x)) \quad .$$

EXERCISE 16. (Rosza Péter) Show that $+$ and L_o can be replaced by $|x-y|$ and $1 \dot{-} (1 \dot{-} L_o(x)))$ in Exercises 14 and 15.

THEOREM 7. (Rosza Péter) There is a recursively defined function which is not p.r.

Proof: Consider the function $\Phi\,(x,y)$ defined recursively by

$$\Phi(o,x) = s(x)$$

$$\Phi(1,x) = L_o(x)$$

$$\Phi(J(0,J(y,z)),x) = \Phi(\cancel{z}^{\,y},\Phi(\cancel{y}^{\,z},x))$$

$$\Phi(J(1,J(y,z)),x) = \Phi(y,x)+\Phi(z,x)$$

$$\Phi(J(2,z),0) = 0$$

$$\Phi(J(2,z)\ ,\ sx) = \Phi(z,\Phi(J(2,z),x))$$

$$\Phi(J(y+3,z),\ x) = 0 \quad .$$

Then $\{\ \Phi(m,x)\ \}_{m=0}^{\infty}$ is an enumeration (with repetions) of all the 1-ary p.r. functors (cf. Exercise 15). Suppose Φ were p.r. Then

$\Phi(x,x) + 1$ is a 1-ary p.r. function and for some m

$$\Phi(m,x) = \Phi(x,x) + 1 \ .$$

However this equation is not valid when x is m . Hence Φ is not p.r.

Our next task is to define a formal system which explicates the arithmetic of p.r. functions and prove the two Gödel Incompleteness Theorems for this system. For this purpose it is convenient to define a system BRA which contains only 1-ary and 2-ary functors and then show that BRA is equivalent in an obvious sense to a similar formal system RA which contains n-ary functors. The system BRA is obtained from Church's Binary Recursive Arithmetic (Journal de Math. Pures et App., vol. 36 (1957), pp. 39-55) by omitting the propositional and function variables.

DEFINITION 7. By BRA we mean the formal system described briefly below.

The primitive symbols of BRA are the numeral constant 0 ; the numerical variables x_0 , x_1 , $x_2, \ldots$; constant functors s , θ , u , v ; functor connectives C and R ; term connective $=$; propositional connectives $\sim$ and $\supset$; and the improper symbols (,) , and ' .

The concepts of 1-ary and 2-ary functor, term, and formula are defined inductively as follows: s , θ , and u are 1-ary functors; v is a 2-ary functor; if g_1 , g_2 are 1-ary functors and h_1 , h_2 are 2-ary functors then $Ch_1 g_1 g_2$ is a 1-ary functor and $Rg_1 h_1 h_2$ is a 2-ary functor; 0 , x_0 , x_1 ,..., are terms; if A and B are terms, g is a 1-ary functor, and h is a 2-ary functor, then $g(A)$ and $h(A,B)$ are terms; if A and B are terms, then $A = B$ is a formula; if P and Q are formulas, then $- P$ and $(P \supset Q)$ are formulas.

The axioms and axiom schema are

0. $\sim s(0) = 0$
1. $\theta(x_0) = 0$
2. $u(x_0) = x_0$
3. $v(x_0,x_1) = x_1$
4. $x_0 = x_1 \supset . x_0 = x_2 \supset x_1 = x_2$
5. $x_0 = x_1 \supset f(x_0) = f(x_1)$
6. $x_0 = x_1 \supset g(x_0,x_2) = g(x_1,x_2)$
7. $x_1 = x_2 \supset g(x_0,x_1) = g(x_0,x_2)$
8. $Cfgh(x_0) = f(g(x_0),h(x_0))$
9. $Rfgh(x_0,0) = f(x)$
10. $Rfgh(x_0,s(x_1)) = g(h(x_0,x_1), Rfgh(x_0,x_1))$
11. $P \supset . Q \supset P$
12. $P \supset (Q \supset P_1) \supset . P \supset Q \supset . P \supset P_1$
13. $\sim P \supset \sim Q \supset . Q \supset P$.

The rules of inference are modus ponens, substitution of terms for numerical variables, and a rule of simple induction —— From $P(0)$ and $P(x_0) \supset P(sx_0)$ to infer $P(x_0)$.

We shall use the following notations in the meta-language: x, y, z, w, y_1, z_1,..... will range over the numerical variables of RA; f, g, h, f_1,..... will range over the functors; A, B, A_1,...... will range over the terms; m, n, m_1,..... will range over the numerals — i.e., 0, $s(0)$, $s(s(0))$,..... ; and P, Q, P_1,..... will range over the formulas; if α is a formula schema, then $\vdash \alpha$ will mean that every instance of α is a theorem of BRA.

EXERCISE 17. Show

$$\vdash x = x$$

$$\vdash x = y \supset y = x$$

$$\vdash x = y \supset . y = z \supset x = z$$

$$\vdash Rfvv(x,y) = f(x)$$

EXERCISE 18. Show that Axiom 5 can be derived from the other axoms (excluding Axiom 7).

EXERCISE 19. Prove a deduction theorem for BRA. Show that the rule of simple induction on any variable (as opposed to just on x_o) and the rule of substitutivity of equality can be derived.

EXERCISE 20. Define a functor σ (addition) for which

$$\vdash \sigma(x,o) = x \quad , \text{ and}$$

$$\vdash \sigma(x,sy) = s(\sigma(x,y)) \quad .$$

Following Church (ibidem) we can define functors representing the pairing functions of Definition 6. We shall use the letters J, K, and L as abbreviations for these functors. Dana Scott has shown that Axiom 7 can be derived from the other axioms. Suppose we call a 2-ary functor g <u>substitutive</u> if

$$y = z \supset g(x,y) = g(x,z)$$

is provable without Axiom 7. Then Scott's proof procedes by finding for each 2-ary $RGH_1 H_2$ with substitutive H_2 , a 1-ary F such that

$$RGH_1 H_2(x,y) = F(J(J(x,y), 0) + y)$$

is provable without using Axiom 7. Since v , + , and J are substitutive, each binary functor is substitutive.

EXERCISE 21.. Define a formal system RA which contains n-ary functors (n= 1,2,3,.....) in a manner analogous to the definition of BRA. Show that BRA and RA are equivalent in an obvious sense.

EXERCISE 22. Show that for each variable - free term (vft) A , there is a numeral n , called the <u>numeral computed from</u> A , such that

$$\vdash A = n \ .$$

Find an algorithm which from A yields a proof of $A = n$ which does not use simple induction and uses substitution only on axioms.

EXERCISE 23. Let BRA' be the formal system obtained from BRA by eliminating substitution, allowing all instances of axioms of BRA as axioms, and modifying the rule of induction to be — From $P(0)$ and $P(y) \supset P(sy)$ to obtain $P(A)$ for any term A . Show that the theorems of BRA and BRA' are identical.

DEFINITION 8. A variable - free formula $A = B$ is called <u>valid</u> iff the same numerals are computed from A and B . If P is a vff , then $\sim P$ is <u>valid</u> iff P is <u>not valid</u> . If $(P \supset Q)$ is a vff , then $(P \supset Q)$ is <u>not valid</u> iff P is <u>valid</u> and Q is <u>not valid</u>. A <u>numerical instance</u> of a formula P is a vff obtained by substituting numerals for all the variables of P . A formula P is <u>valid</u> if every numerical instance of P is valid.

THEOREM 8. Every theorem P of BRA is valid.

Proof: The demonstration procedes by a meta-induction on the length of the proof of P . The details are left to the reader.

Since a formula and its negation can not both be valid, we have as an immediate consequence of the last theorem,

THEOREM 9. BRA is consistent. I.e., there is no formula P such both P and $\sim P$ are theorems.

Theorem 8 has a partial converse. Namely

THEOREM 10. Every valid vff is a theorem of BRA.

Proof: Use Exercise 22, an analogue of Exercise 6, and Theorem 8.

We shall make use of the Godel Incompleteness Theorems in classifying extensions of recursive arithmetic. The Gödel numbering and the statement and proof of the first Gödel theorem for BRA given below are essentially those of Church (Logic Seminar, Princeton University, 1958-59). The treatment of the second Gödel Theorem stems in part from the same source.

DEFINITION 9. To each functor f of BRA we inductively assign a unique numeral "f" , called the Gödel number of f , by the following rules:

1) "s" = 0, "[illegible]" = 1, "u" = 2, "v" = 0,

2) "Cfgh" = J("f",J("g","h"))+3, and

3) "Rfgh" = J("f",J("g","h"))+1 .

DEFINITION 10. To each term A we inductively assign a unique numeral "A", called the Gödel number of A, by the following rules:

1) "0" = 0,

2) "x_i" = 3(i+1), for i = 0, 1, 2,....,

3) "f(A)" = 3J("f","A")+1, and

4) "f(A,B)" = 3J("f",J("A","B"))+2 .

DEFINITION 11. To each formula P we inductively assign a unique numeral "P", called the Gödel number of P, by the following rules:

1) "A = B" = 3J("A","B"),

2) "P $\supset$ Q" = 3J("P","Q")+1, and

3) "$\sim$ P" = 3("P")+2 .

Notice that each numeral is assigned to exactly one singulary functor, one binary functor, one term, and one formula. Following Gödel, we can define p.r. functions which mirror, via the Gödel numbering given in Definitions 9, 10, and 11, many of the syntactical operations of BRA

EXERCISE 24. Define functors mp , sbt , sbf , sb , ind , ax , num , and sub satisfying [1] - [8]:

[1] $mp("P","P \supset Q") = "Q"$.

[2] $sbt(p_0^{"A_0"} \ldots\ldots p_K^{"A_K"}, "B") = "S^{x_0 \ldots\ldots x_K \; x_{K+1} \; x_{K+2} \ldots\ldots}_{A_0 \qquad A_K \;\; 0 \quad\; 0 \;\; \ldots\ldots} B|"$.

[3] $sbf(p_0^{"A_0"} \ldots\ldots p_K^{"A_K"}, "P") = "S^{x_0 \ldots\ldots x_K \; x_{K+1} \; x_{K+2} \ldots\ldots}_{A_0 \qquad A_K \;\; 0 \quad\; 0 \;\; \ldots\ldots} P|"$.

[4] $sb(J("A", n), "P") = "S_A^{x_n} P|"$

[5] $ind("P(0)","P(x_0) \supset P(sx_0)") = "P(x_0)"$.

[6] $ax(0) = "s(0) \neq 0"$,

$ax(1) = "o(x_0) = 0"$,

$ax(2) = "u(x_0) = x_0"$,

$ax(3) = "v(x_0,x_1) = x_1"$,

$ax(4) = "x_0 = x_1 \supset . x_0 = x_2 \supset x_1 = x_2"$,

$ax(9"f"+5) = "x_0 = x_1 \supset f(x_0) = f(x_1)"$

$ax(9"g"+6) = "x_0 = x_1 \supset g(x_0,x_2) = g(x_1,x_2)"$;

$ax(9"g"+7) = "x_1 = x_2 \supset g(x_0,x_1) = g(x_0,x_2)"$,

$ax(9("Cfgh" \dot{-} 3)+8) = "Cfgh(x_0) = f(g(x_0),h(x_0))"$,

$ax(9"Rfgh") = "Rfgh(x_0,0) = f(x_0)"$,

$ax(9"Rfgh"+1) = "Rfgh(x_0,sx_1) = g(h(x_0,x_1),Rfgh(x_0,x_1))"$,

$ax(9J("P","Q")+11) = "P \supset . Q \supset P"$,

$ax(9J("P",J("Q","P_1"))+12)$

$= "P \supset [Q \supset P_1] \supset . P \supset Q \supset . P \supset P_1"$.

$ax(9J("P","Q")+13) = "\sim P \supset \sim Q \supset . \; Q \supset P"$.

[7] $num(n)$ is the Godel number of the numeral $\underbrace{ss\ldots\ldots s0}_{n}$.

[8] $sub(z,"P") = "S^{X_0}_{\underbrace{ss\ldots\ldots s0}_{z}} P|"$.

DEFINITION ~~10~~ 12. Let

$$\mathrm{th}(4y) = \mathrm{ax}(y),$$

Ky

$$\mathrm{th}(4y+1) = \mathrm{sb}(KKy, LKy, \mathrm{th}(Ly)),$$

$$\mathrm{th}(4y+2) = \mathrm{mp}(\mathrm{th}(Ky), \mathrm{th}(Ly)),$$

$$\mathrm{th}(4y+3) = \mathrm{ind}(\mathrm{th}(Ky), \mathrm{th}(Ly)) \quad .$$

Then th(0), th(1), th(2),....... enumerates the Gödel numbers of all formulas which can be obtained from the axioms by modus ponens, substitution, and simple induction on x_0. Hence th(0), th(1),..... enumerates the Gödel numbers of all and only the theorems of BRA.

THEOREM 11. (Gödel's Incompleteness Theorem for BRA.)
Not every valid formula of BRA is a theorem. In particular let i be a numeral such that

$$i = \text{"}\mathrm{th}(x_1) \neq \mathrm{sub}(x_0, x_0)\text{"} \quad .$$

Then $\mathrm{th}(x_1) \neq \mathrm{sub}(i,i)$ is valid but unprovable.

Proof: Let j be the mumberal such that $j = \text{"}\mathrm{th}(x_1) \neq \mathrm{sub}(i,i)\text{"}$.
Then

1) $j = \mathrm{sub}(i,i)$

is valid and hence provable (since it contains no variables).

Now suppose that

2) $\mathrm{th}(x_1) \neq \mathrm{sub}(i,i)$

is provable. Then there exists a numeral k such that

$$\mathrm{th}(k) = \text{"}\mathrm{th}(x_1) \neq \mathrm{sub}(i,i,)\text{"} \text{ — i.e.,}$$

3) $\mathrm{th}(k) = j$.

From 1) and 3) we obtain $\mathrm{th}(k) = \mathrm{sub}(i,i)$; from 2) we obtain $\mathrm{th}(k) \neq \mathrm{sub}(i,i)$ by substitution. But this contradicts the consistency of BRA (cf.Theorem 9). Hence $\mathrm{th}(x_1) \neq \mathrm{sub}(i,i)$ is unprovable.

On the other hand, since $th(x_1) \neq sub(i,i)$ is unprovable, we have for each n that $th(n) \neq j$ is valid. Hence $th(x_1) \neq sub(i,i)$ is valid.

DEFINITION 12. Suppose that ξ is a term (formula) in which the terms $A_1, \ldots\ldots, A_n$ appear underlinded. Then by "ξ" we shall mean a term of BRA which is built up inductively as in Definition 10 (Def.11) with the exception that $num(A_i)$ is used instead of "A_i" . Hence "ξ" is a term in just those variables which appear in $A_1, \ldots\ldots, A_n$.

E.g., $\text{"}f\underline{x} = \underline{g(hx,0)}\text{"} \rightsquigarrow 3J(3J(\text{"}f\text{"},numx)+1,num\ g(hx,0))$.

Note that $\text{"}s\underline{x}\text{"} = \text{"}\underline{sx}\text{"}$, since $3J(\text{"}s\text{"},numx)+1 = num(sx)$.

THEOREM 12. For every singulary functor f there exists a singulary functor Df such that $th(Df(x)) = \text{"}f\underline{x} = \underline{fx}\text{"}$. Similarly for every binary functor g there exists a binary functor Dg such that $th(Dg(x,y)) = \text{"}g(\underline{x},\underline{y}) = \underline{g(x,y)}\text{"}$.

Proof: The proof procedes by a meta-induction on the length of the definition of a functor. The details are left to the reader.

As a corollary to Theorem 12 we have

THEOREM 13. For each singulary f we can prove

$$f(x) = y \supset th(Df(x)) = \text{"}f\underline{x} = y\text{"} \ .$$

An analogous result holds for binary functors.

Proof:

$$f(x) = y \vdash th(Df(x)) = 3J(3J(\text{"}f\text{"},num\ x)+1,num(f(x))), \text{ by Theorem 12}$$
$$= 3J(3J(\text{"}f\text{"},numx)+1,num\ y)) = \text{"}f\underline{x} = y\text{"} \ .$$

THEOREM 14. (Gödel's Second Theorem for BRA). The formula $th(y) \neq \text{"}0 = 1\text{"}$ is valid but unprovable in BRA.

Proof: We shall formalize the proof of Theorem 11. Let i and j be the numerals defined in Theorem 11. Then

1) $th(x) = j \supset th(Dth(x)) = \text{"}th(\underline{x}) = j\text{"}$, by Th 13, and

2) $th(\cancel{sub}\,Dsub(i,i)) = \text{"}sub(i,i) = j\text{"}$, by Th 13.

Let t be a numeral such that

$$th(t) = \text{"}x_0 = x_2 \supset . \; x_1 = x_2 \supset x_0 = x_1\text{"} .$$

Let $f(x) = 4J[J(\text{"}th(\underline{x})\text{"},0),4J[J(\text{"}sub(i,i)\text{"},1), 4J[J(num\ j,2),t]+1]+1]+1$.

Then $th(fx) = \text{"}th(\underline{x}) = j \supset . \; sub(i,i) = j \supset th(\underline{x}) = sub(i,i)\text{"}$.

Let $g(x) = 4J(Dsub(i,i),4J(Dth(x),f(x))+2)+2$. Then

3) $th(x) = j \supset th(gx) = \text{"}th(\underline{x}) = sub(i,i)\text{"}$, by 1), 2) .

By the definition of j ,

4) $th(x) = j \supset th(4J(J(num\ x,1),x)+1) = \text{"}th(\underline{x}) \neq sub(i,i)\text{"}$.

Let t' be a numeral such that

$$th(t') = \text{"}x_0 = x_1 \supset . \; x_0 \neq x_1 \supset 0 = 1\text{"} .$$

Let $h(x) = 4J(J(\text{"}th(x)\text{"},0),4J(J(\text{"}sub(i,i)\text{"},1),t')+1)+1$.

Then by 3) and 4) we have

5) $th(x) = j \supset$

$$th(4J[4J(J(num\ x,1),x)+1, 4J(gx,hx)+2]+2) = \text{"}0 = 1\text{"} .$$

Hence if $th(y) \neq \text{"}0 = 1\text{"}$ were provable, then by substitution and 5) we would have that $th(x) \neq j$ would be provable. I.e., $th(x) \neq sub(i,i)$ would be provable contrary to Theorem 11.

Since $0 = 1$ is not a theorem, $th(n) \neq \text{"}0 = 1\text{"}$ is valid for each n . I.e., $th(y) \neq \text{"}0 = 1\text{"}$ is valid.

THEOREM 15. Let P be any formula which is not a theorem of BRA. Then $th(y) \neq \text{"}P\text{"}$ is valid but unprovable.

Proof: This proof is identical with the last proof except that we take t' to be a numeral such that

$$th(t') = \text{"}x_0 = x_1 \supset . \; x_0 \neq x_1 \supset P\text{"} .$$

By examining the proof of Theorem 14 we obtain

THEOREM 16. The formula $th(y) \neq 3th(x)+2$ is valid but unprovable.

Let us now prove two results for BRA which are analogous to classical results (Tarski's Theorem).

THEOREM 17. There is no p.r. valuation function for the Gödel numbering given above. I.e., there is no p.r. functor V such that

$$\vdash V(\text{"}A\text{"}) = A \text{ , for every vft } A.$$

Proof: Let us enumerate the terms which contain no variables except possibly x_o . E.g. let

$A_o(x_o)$ be 0,

$A_1(x_o)$ be x_o,

$A_{2m+2}(x_o)$ be $f(A_{Lm}(x_o))$

where "f" = K(m), and let

$A_{2m+3}(x_o)$ be $g(A_{KLm}(x_o), A_{LLm}(x_o))$

where "g" = K(m) .

It is straight-forward to define a p.r. F such that

$$F(m,n) = \text{"}A_m(n)\text{"}$$

Now suppose there is a p.r. V such that

$$\vdash V(\text{"}A\text{"}) = A, \text{ for every vft } A .$$

Then

$$\vdash V(F(m,n)) = A_m(n) \text{ , for each } m , n .$$

However $V(F(x_o,x_o)) + 1$ would be a p.r. term in x_o .

Hence $V(F(x_o,x_o)) + 1$ is $A_r(x_o)$ for some r. Now

$$\vdash V(F(r,r)) = A_r(r) = V(F(r,r)) + 1$$

which is clearly a contradiction.

In the same spirit we can prove the following result, the details of which are left to the reader.

THEOREM 18. There is no p.r. truth formula for the Gödel numbering given above. I.e., there is no p.r. formula $T(x_o)$ such that

$$\vdash T("P") \equiv P \; ; \quad \text{for every vff } P \; .$$

DEFINITION 14. Let BRA_o be the formal system whose axioms are all the variable-free instances of the axioms of BRA and whose only rule of inference is modus ponens. Let ax_o be a p.r. functor which enumerates the Gödel numbers of the axioms of BRA. Let

$$th_o(2y) = ax_o(y) \; , \quad \text{and}$$

$$th_o(2y+1) = mp(th_o(Ky) \; , \; th_o(Ly)) \; .$$

Then $th_o(0)$, $th_o(1)$, ... enumerates the Gödel numbers of all the theorems of BRA_o .

OPEN PROBLEM. The formula

$$th_o(y) \neq "0 = 1"$$

is valid but conjectured by this writer to be unprovable in BRA. If this is indeed the case then this is an interesting example counter to the usual situation where induction-free fragments are usually provably consistent within the original system.